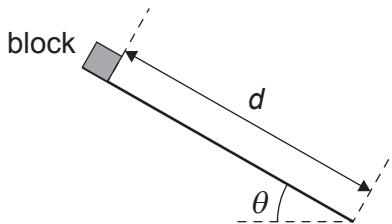


PAPER 1 / QUESTION 118

5 November 2025 / TZ1 (file label) / HL Paper 1A / original question 5 / 8825-6707

- 118.** A block of mass m is released from rest and slides down a ramp of length d that makes an angle θ with the horizontal. A constant frictional force F acts on the block.



What is the kinetic energy of the block at the bottom of the ramp?

- A. $mgd - Fd \cos \theta$
- B. $mgd - Fd$
- C. $mgd \sin \theta - Fd \cos \theta$
- D. $mgd \sin \theta - Fd$